Time Series Analysis of Amazon (AMZN) Stock Price using MATLAB (data from finance.yahoo.com)

Amazon.com is a multinational technology and e-commerce company based in Seattle, Washington. It was founded by Jeff Bezos in 1994 and is one of the largest companies in the world. Its operations include online retail, cloud computing, artificial intelligence, and logistics. Its stock is listed on the NASDAQ under the symbol AMZN..

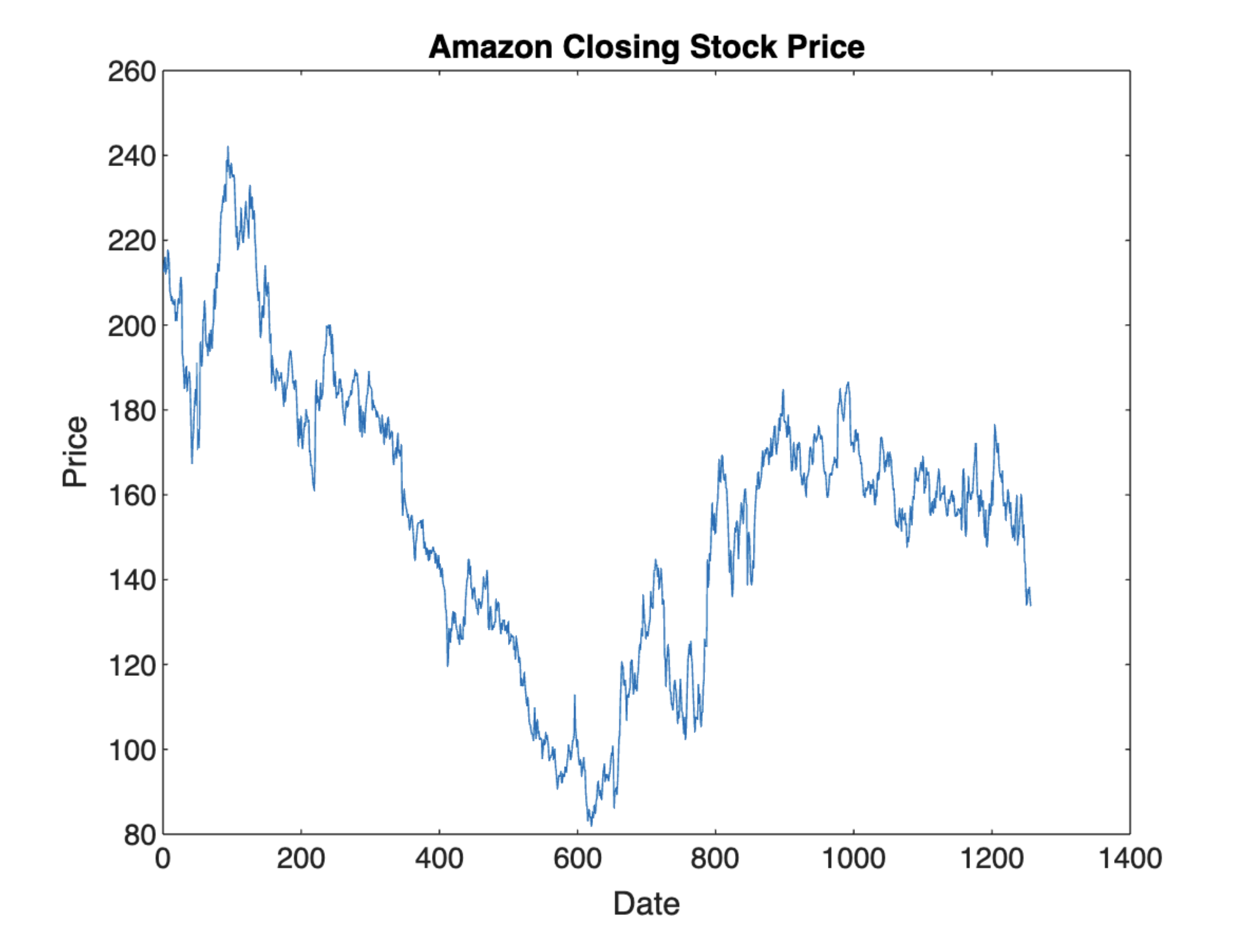
I picked Amazon as it is popular and widely followed stock. My goal is to apply time series techniques to understand Amazon stock price movements and effectiveness of various forecasting models. I used data for the last five years from April 19, 2020, to April 19, 2025 using Yahoo Finance.

I used a range of time series techniques, including ARMA, ARIMA and GARCH models, to understand Amazon stock returns over the five year period. Model performance is evaluated based on the model fit.

Time Series:

Code:

```
figure;
plot(ts.Time, ts.Data);
title('Amazon Closing Stock Price');
xlabel('Date'); ylabel('Price');
```



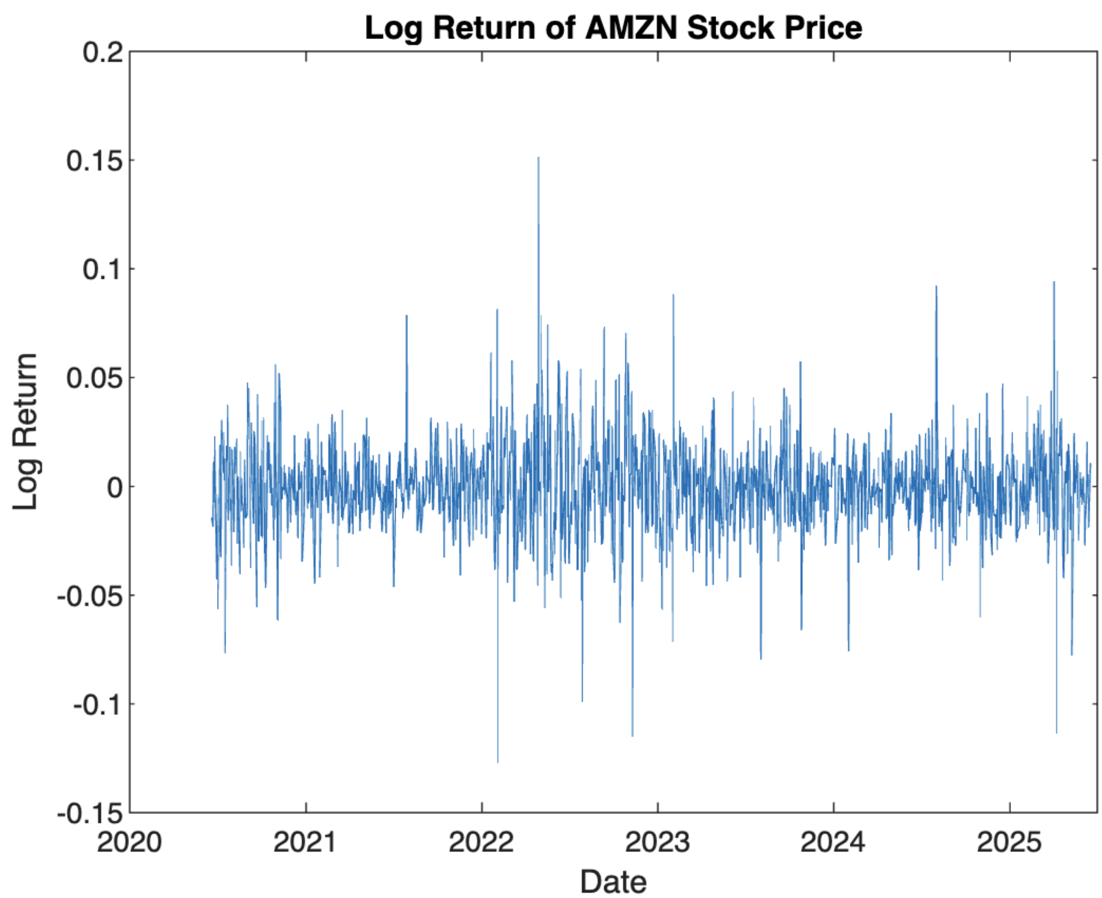
Log Return:

Code:

```
logReturns = diff(log(timeSeriesData));
```

```
figure;
plot(data.Date(2:end), logReturns);
title('Log Returns of Amazon');
xlabel('Date'); ylabel('Log Return');
```

Log Return:



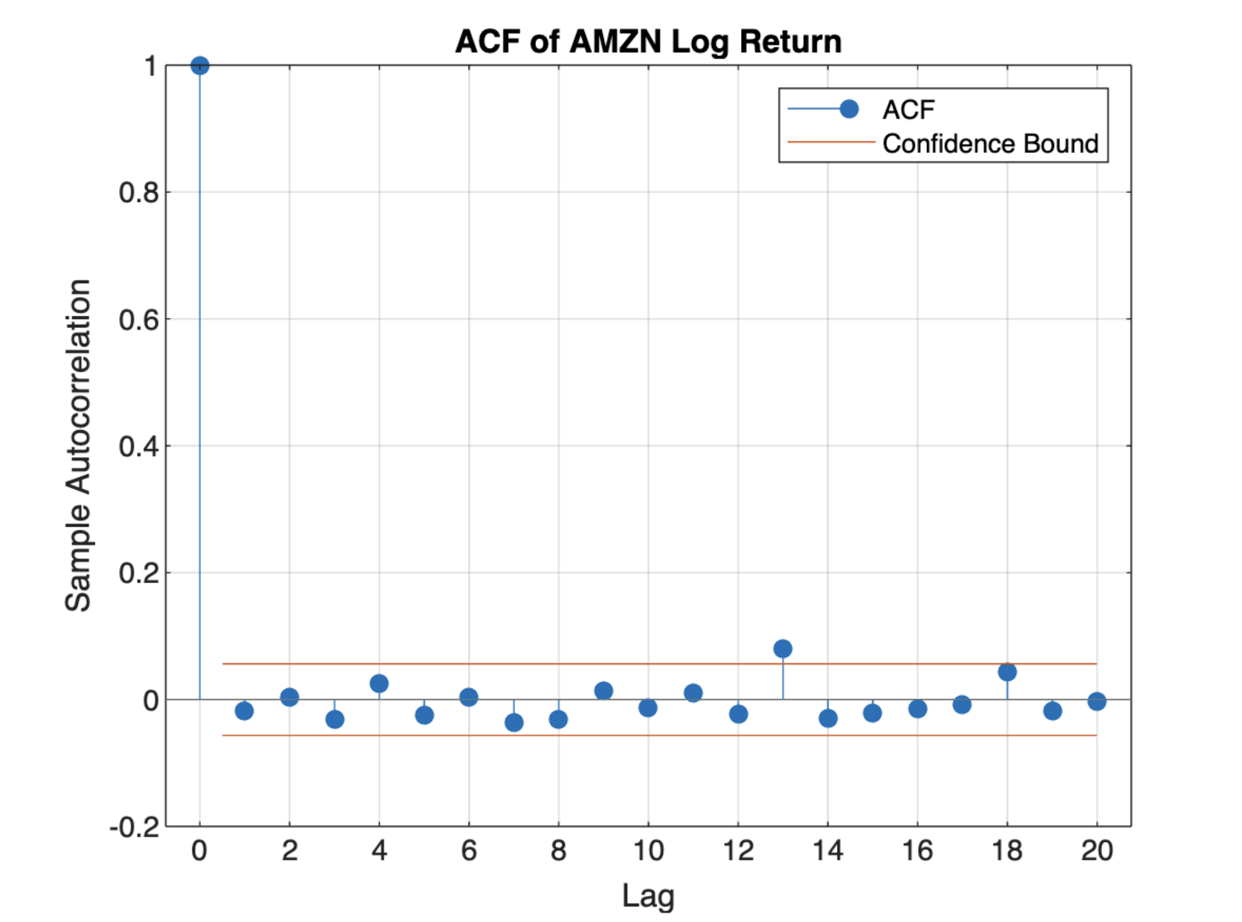
ACF Graph:

Code:

```
figure;
```

```
autocorr(logReturns);
title('ACF of AMZN Log Returns');
```

ACF of AMZN Log Returns:



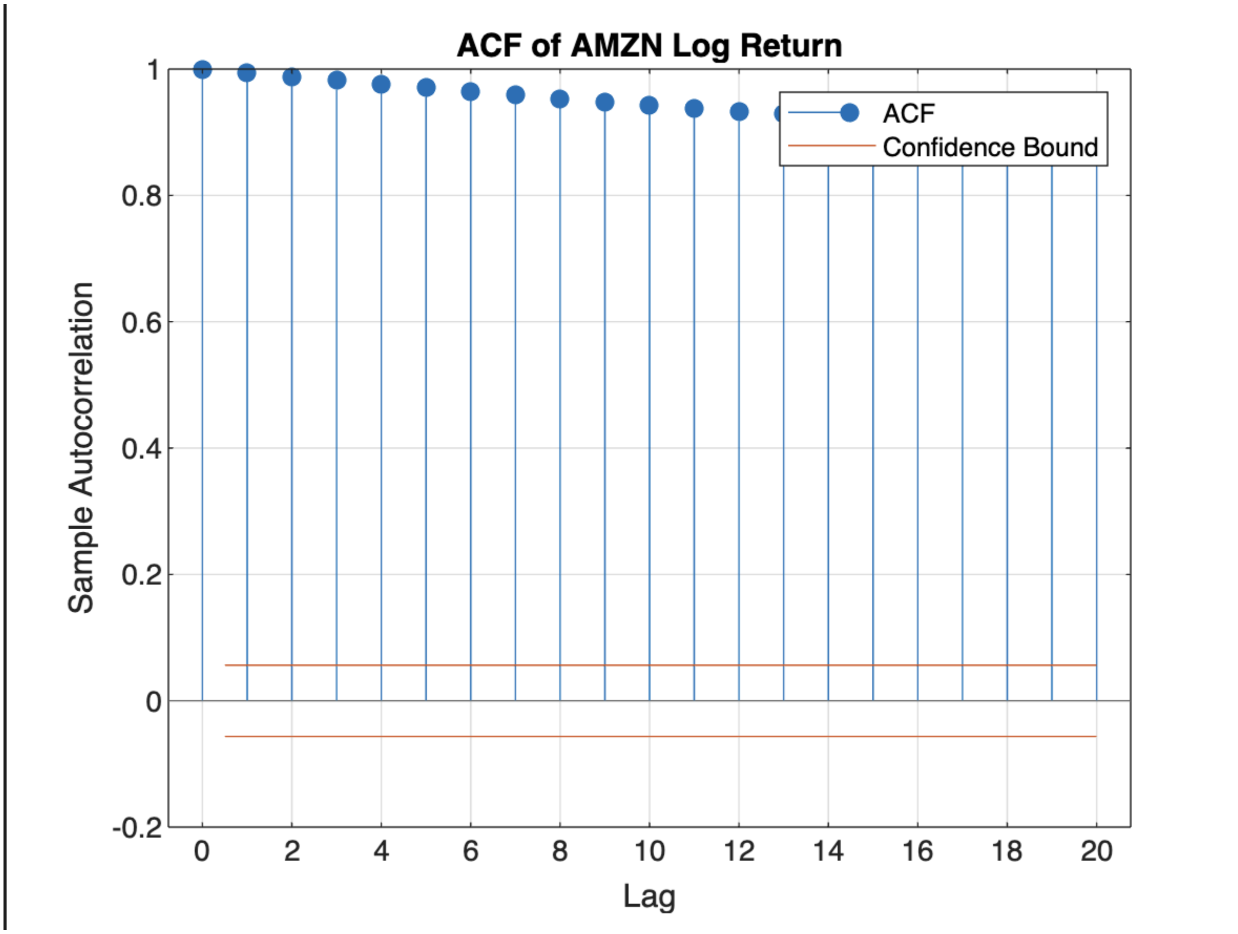
Stationarity (Dickey-Fuller Test):

"Stationary? "   "true"   ", p-value: "   "0.001"

Code(Check for stationarity):

```
[h, p] = adftest(logReturns);
disp(['Stationary? ', string(h==1), ', p-value: ', num2str(p)]);
```

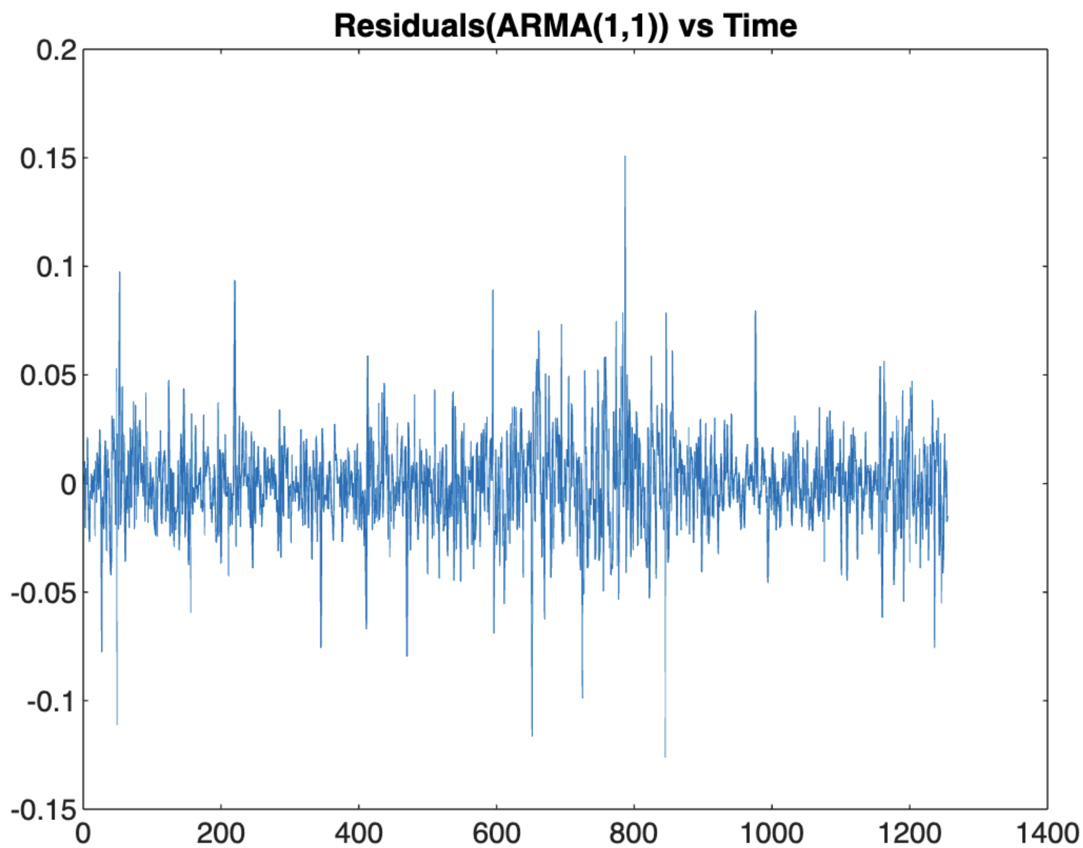
Seasonality from ACF of AMZN Stock Price:



Code:

```
figure;
autocorr(timeSeriesData);
title('ACF of Amazon Stock Price');
```

Residuals of ARMA(1,1) Model:



Code:

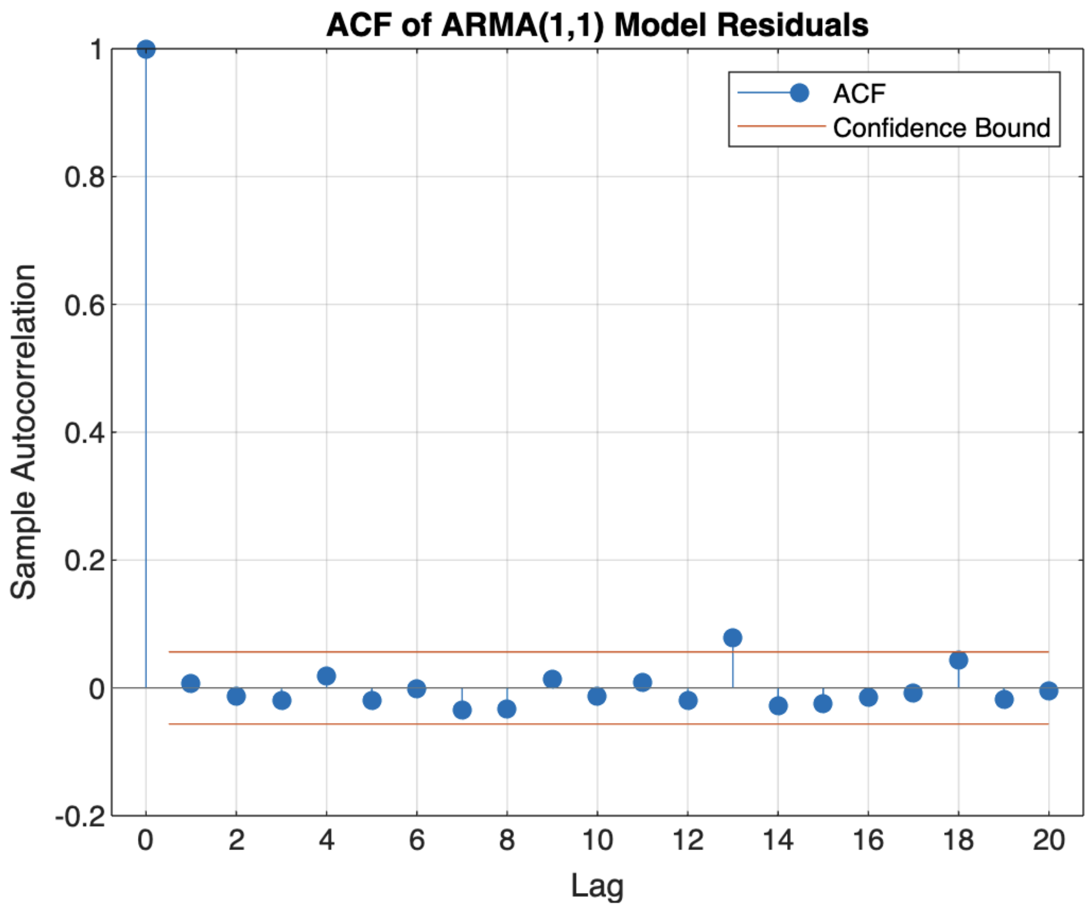
```
figure;
autocorr(timeSeriesData);
title('ACF of Amazon Stock Price');
>> model = arima(1,0,1);
fitModel = estimate(model, logReturns);

% Get residuals
[res,~,logL] = infer(fitModel, logReturns);

figure;
plot(res);
title('Residuals of ARMA(1,1) Model');
```

Results:

ARIMA(1,0,1) Model (Gaussian Distribution):

	Value	StandardError	TStatistic	PValue
Constant	-0.00061468	0.0010531	-0.5837	0.55942
AR{1}	-0.67805	0.37628	-1.802	0.07155
MA{1}	0.65356	0.38802	1.6843	0.092117
Variance	0.00050471	1.1263e-05	44.811	0

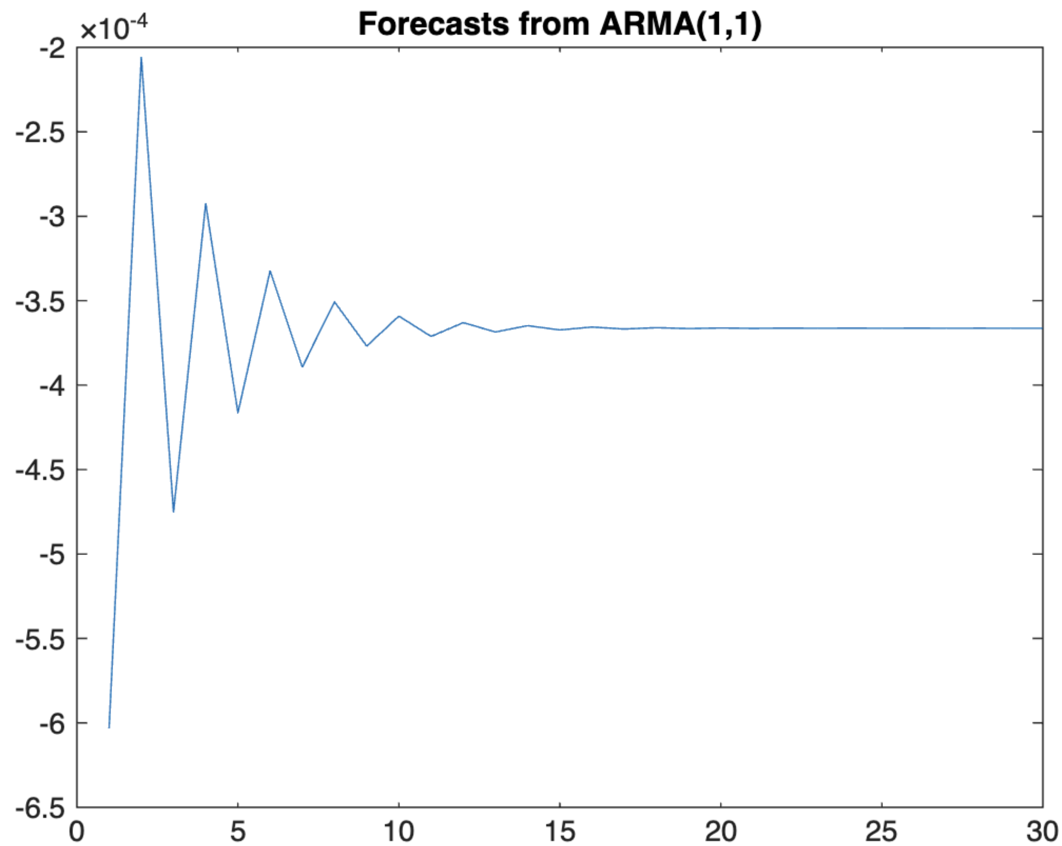
ACF of ARMA(1,1) Residuals:



Code:

```
figure;
autocorr(res);
title('ACF of ARMA(1,1) Residuals');
```

30-Day Forecast Using ARMA(1,1):



Code:

```
figure;
autocorr(res);
title('ACF of ARMA(1,1) Residuals');
>> [yF, ~] = forecast(fitModel, 30, 'Y0', logReturns);
figure;
plot(yF);
title('30-Day Forecast using ARMA(1,1)');
```

GARCH Model:

GARCH(1,1) Conditional Variance Model (Gaussian Distribution):

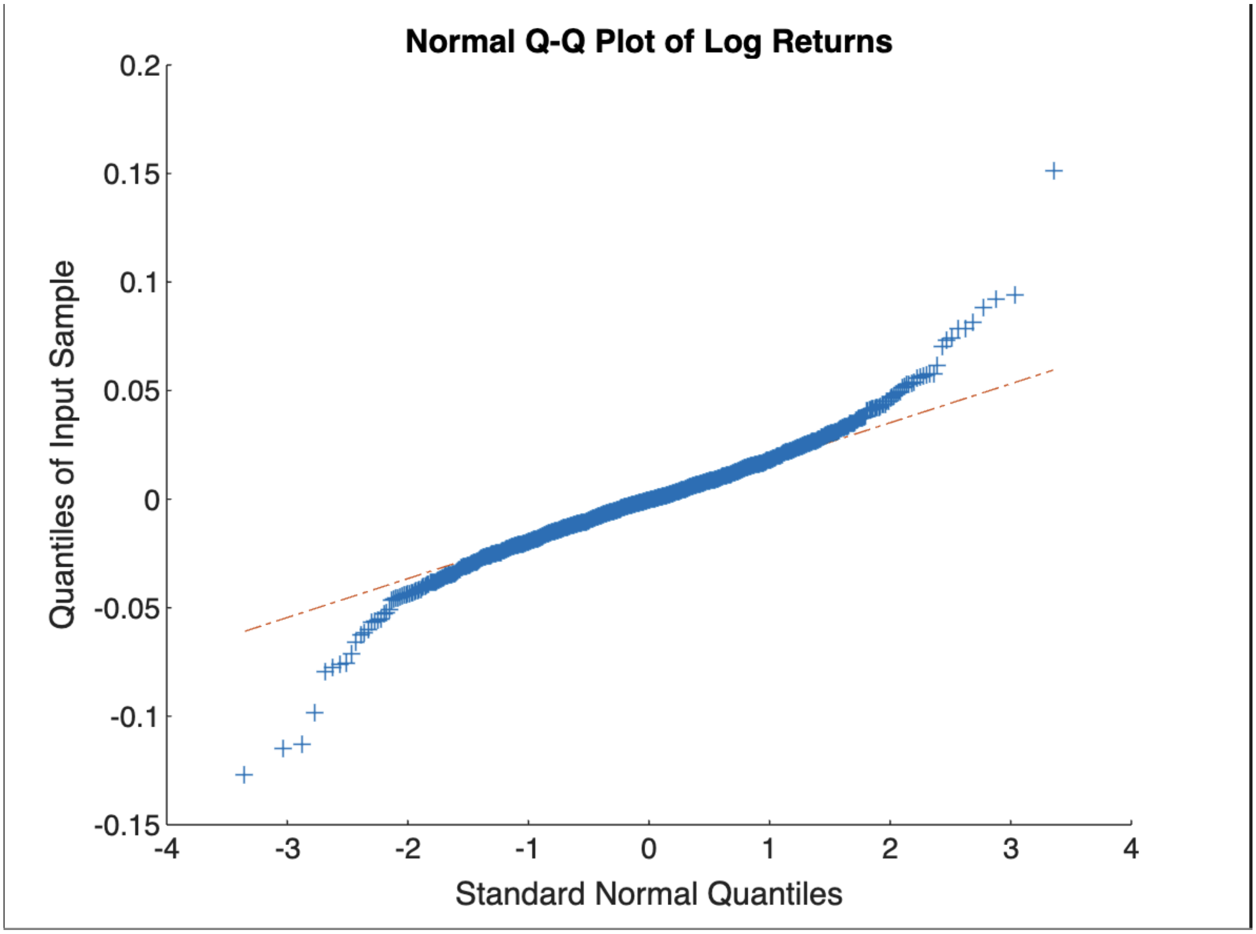
	Value	StandardError	TStatistic	PValue
Constant	1.8116e-05	3.2151e-06	5.6346	1.7543e-08
GARCH{1}	0.90034	0.014208	63.367	0

ARCH{1}	0.065379	0.010038	6.5131	7.3614e-11

Code:

```
garchMdl = garch(1,1);
garchFit = estimate(garchMdl, logReturns);
```

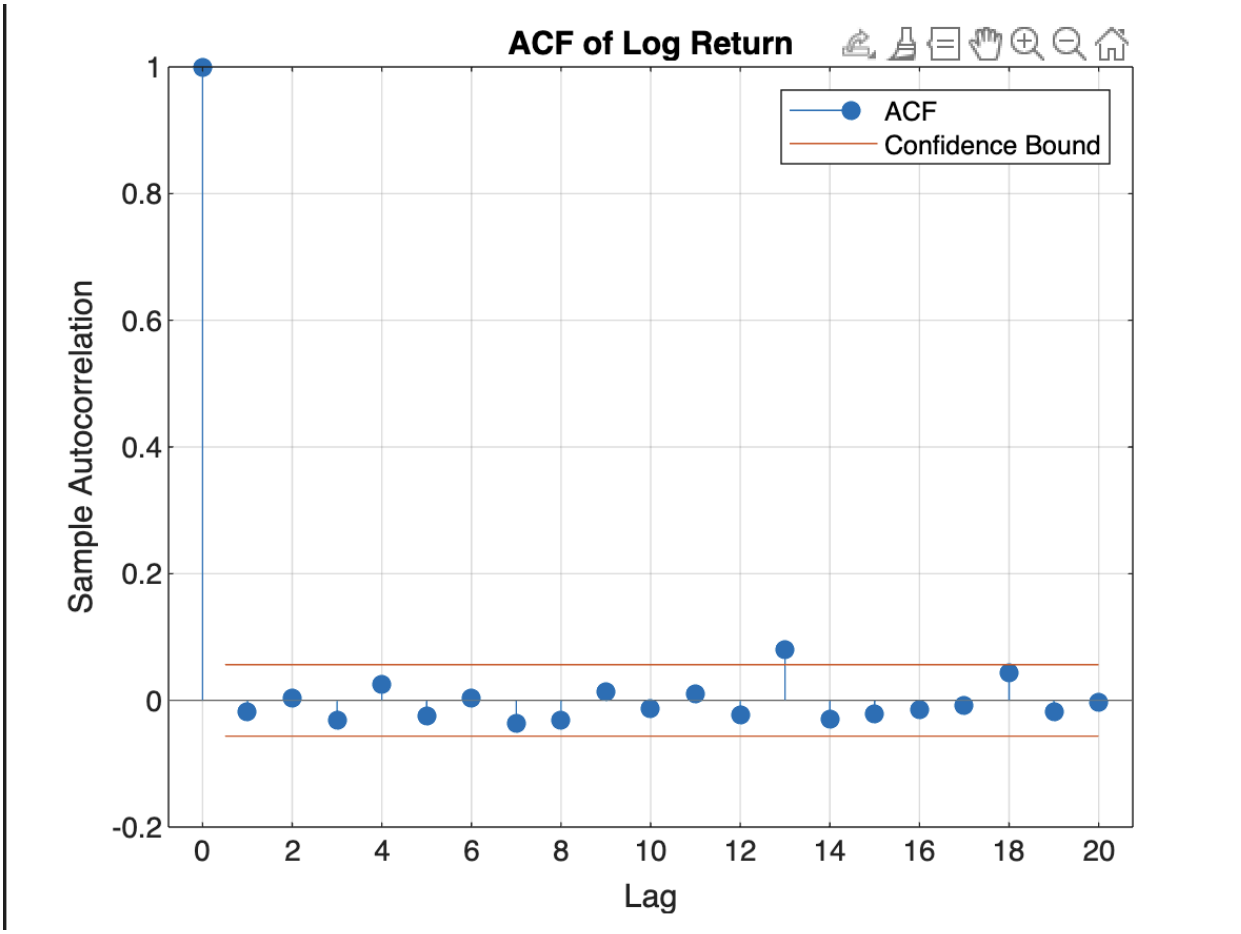
Normal Q-Q Plot of Log Returns:



Code:

```
figure;
qqplot(logReturns);
title('Normal Q-Q Plot of Log Returns');
```

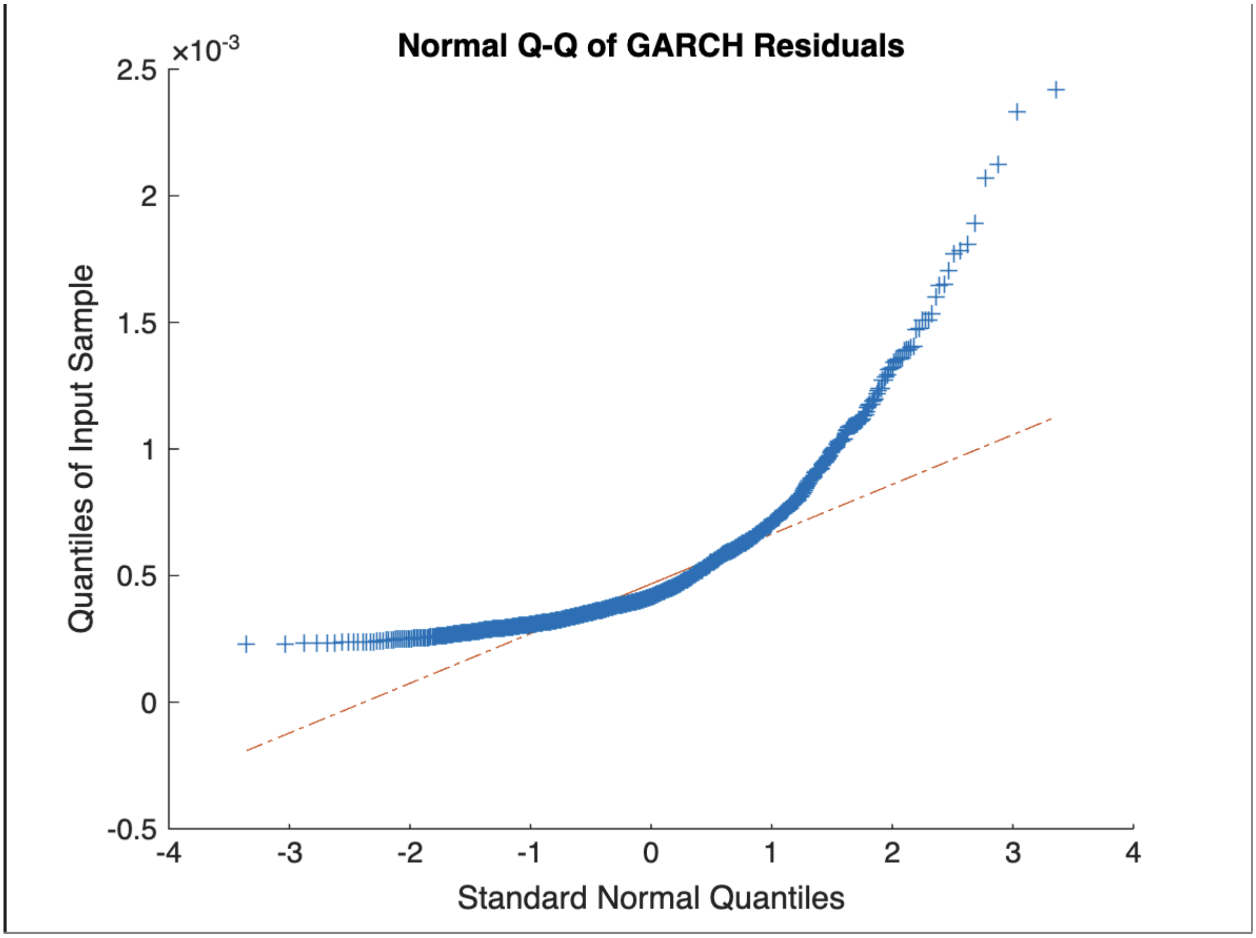
ACF of Log Returns:



Code:

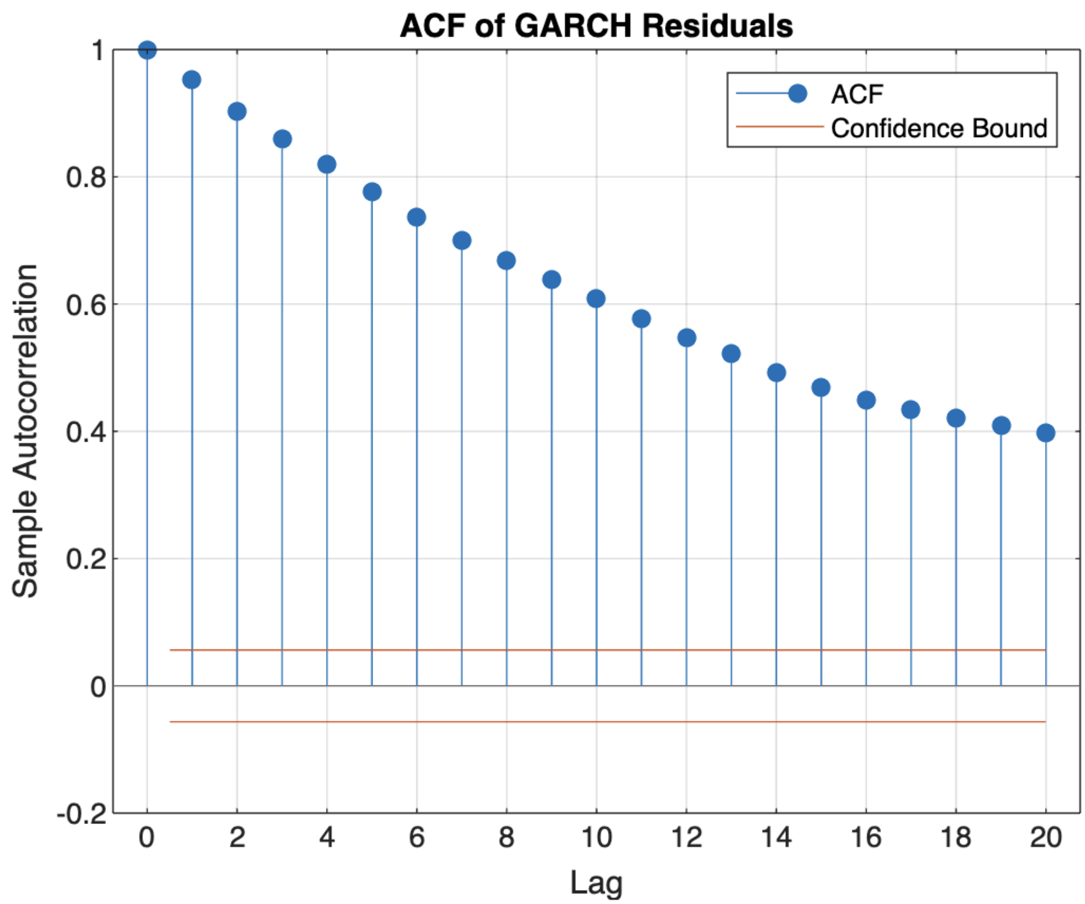
```
figure;
autocorr(logReturns);
title('ACF of Log Returns');
```

Normal Q-Q of GARCH Residuals:



Code:
```
[garchRes, ~] = infer(garchFit, logReturns);
figure;
qqplot(garchRes);
title('Normal Q-Q of GARCH Residuals');
```

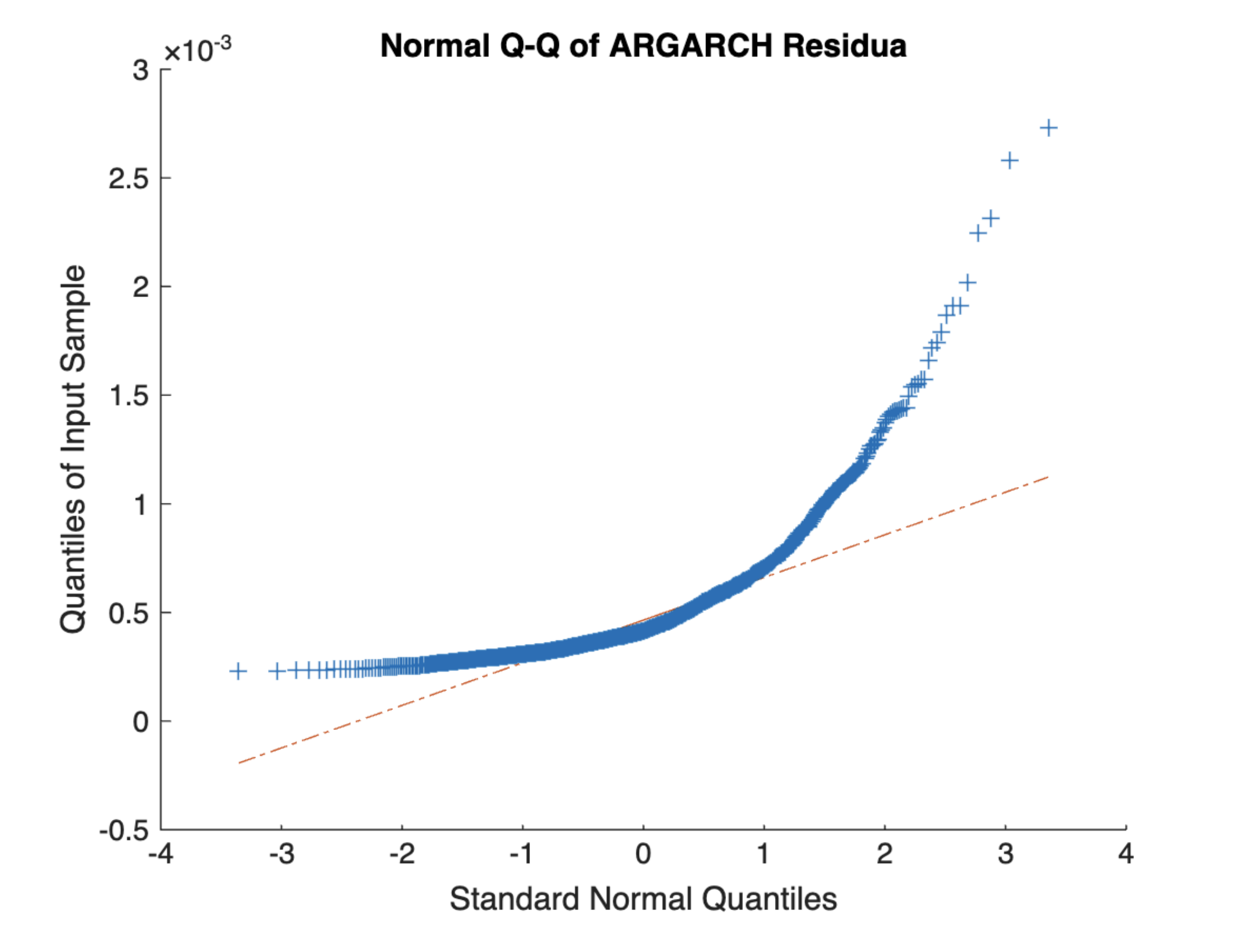
ACF of GARCH Residuals:



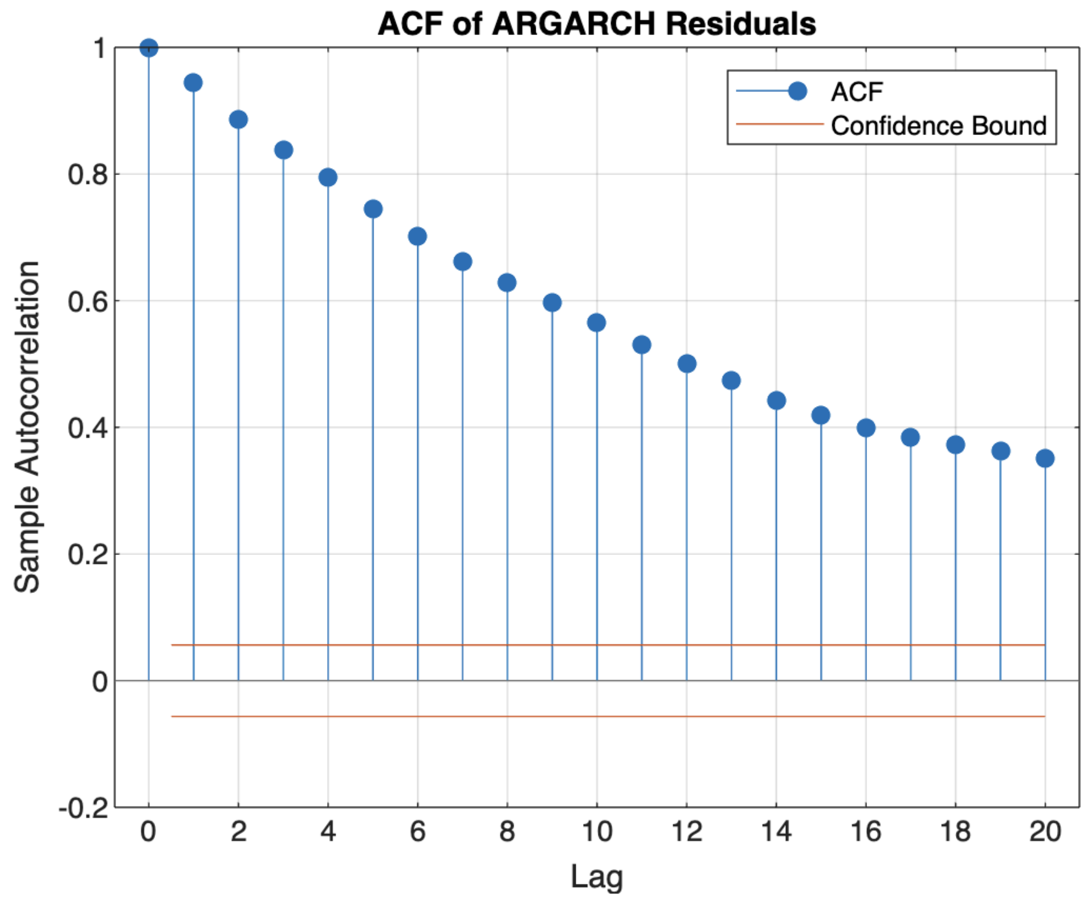
Code:

```
figure;
autocorr(garchRes);
title('ACF of GARCH Residuals');
```

Asymmetric GARCH (e.g., GJR-GARCH):



Normal Q-Q of ARGARCH Residua
×10^-3
3
2.5
2
1.5
1
0.5
0
-0.5
Quantiles of Input Sample
-4
-3
-2
-1
0
1
2
3
4
Standard Normal Quantiles



Code:

```
gjrMdl = gjr(1,1);
gjrFit = estimate(gjrMdl, logReturns);
[gjrRes, ~] = infer(gjrFit, logReturns);

% QQ Plot
figure;
qqplot(gjrRes);
title('Normal Q-Q of GJR-GARCH Residuals');

% ACF
figure;
autocorr(gjrRes);
title('ACF of GJR-GARCH Residuals');
```

GJR(1,1) Conditional Variance Model (Gaussian Distribution):

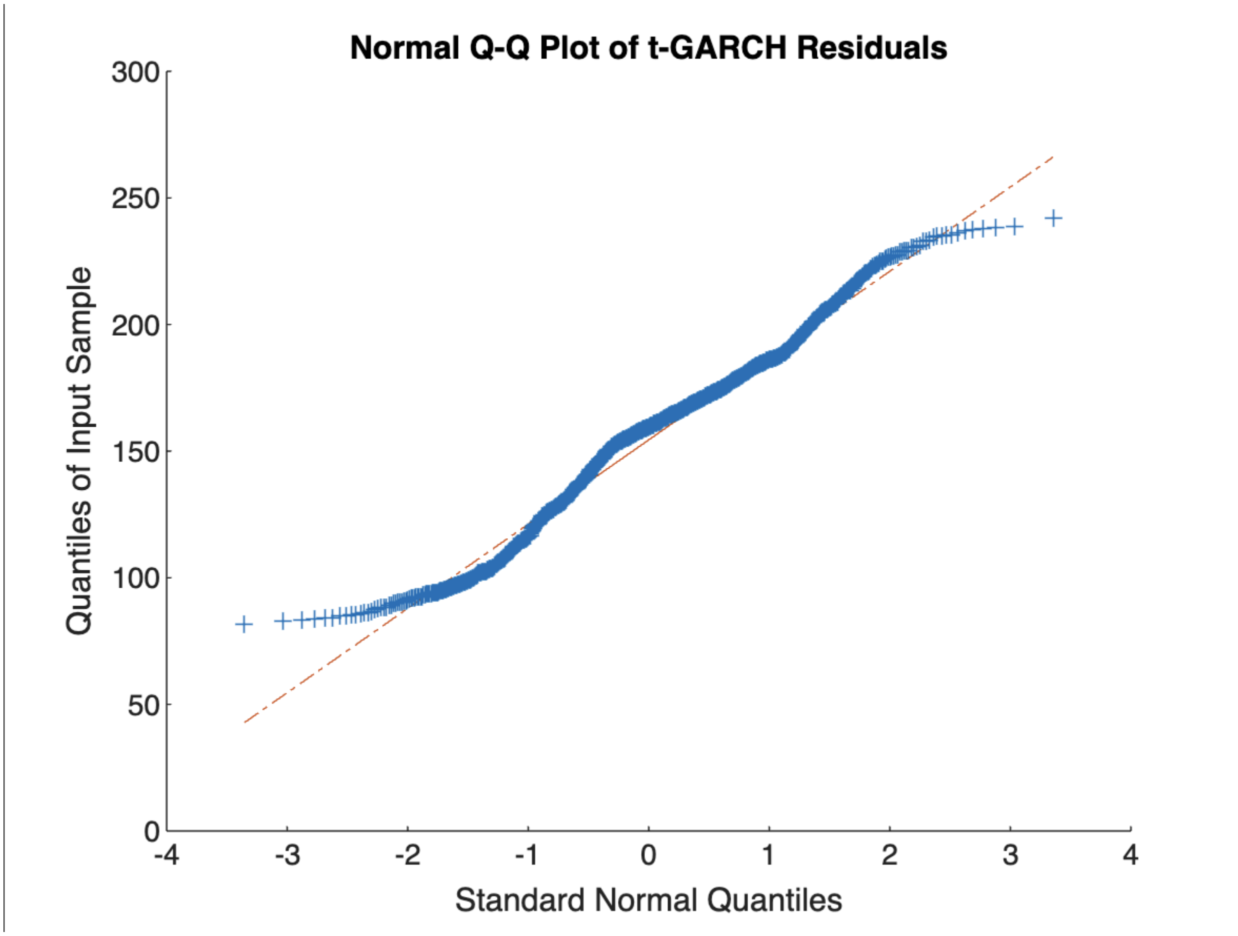
	Value	StandardError	TStatistic	PValue
Constant	2.1494e-05	3.9236e-06	5.4782	4.2957e-08
GARCH{1}	0.88608	0.016423	53.955	0
ARCH{1}	0.078535	0.015754	4.985	6.1966e-07
Leverage{1}	-0.0097798	0.019265	-0.50764	0.6117

GARCH with t-distribution:

GARCH(1,1) Conditional Variance Model (t Distribution):

	Value	StandardError	TStatistic	PValue
Constant	9.2084e-06	4.0764e-06	2.2589	0.023887
GARCH{1}	0.92595	0.018924	48.929	0
ARCH{1}	0.058088	0.015275	3.8028	0.0001431
DoF	4.9725	0.62842	7.9127	2.5181e-15

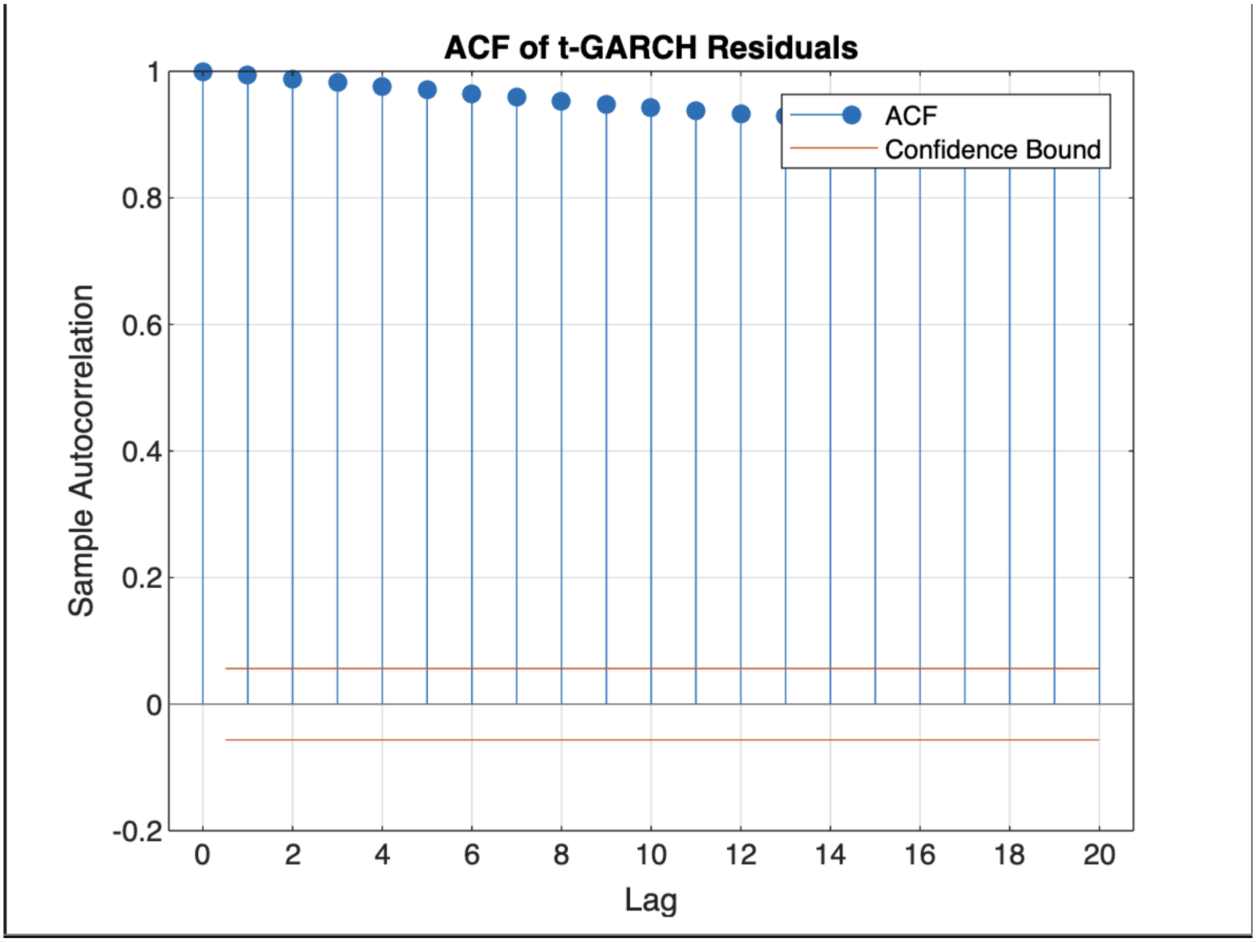
Normal Q-Q Plot of T-GARCH Residual:



Code:

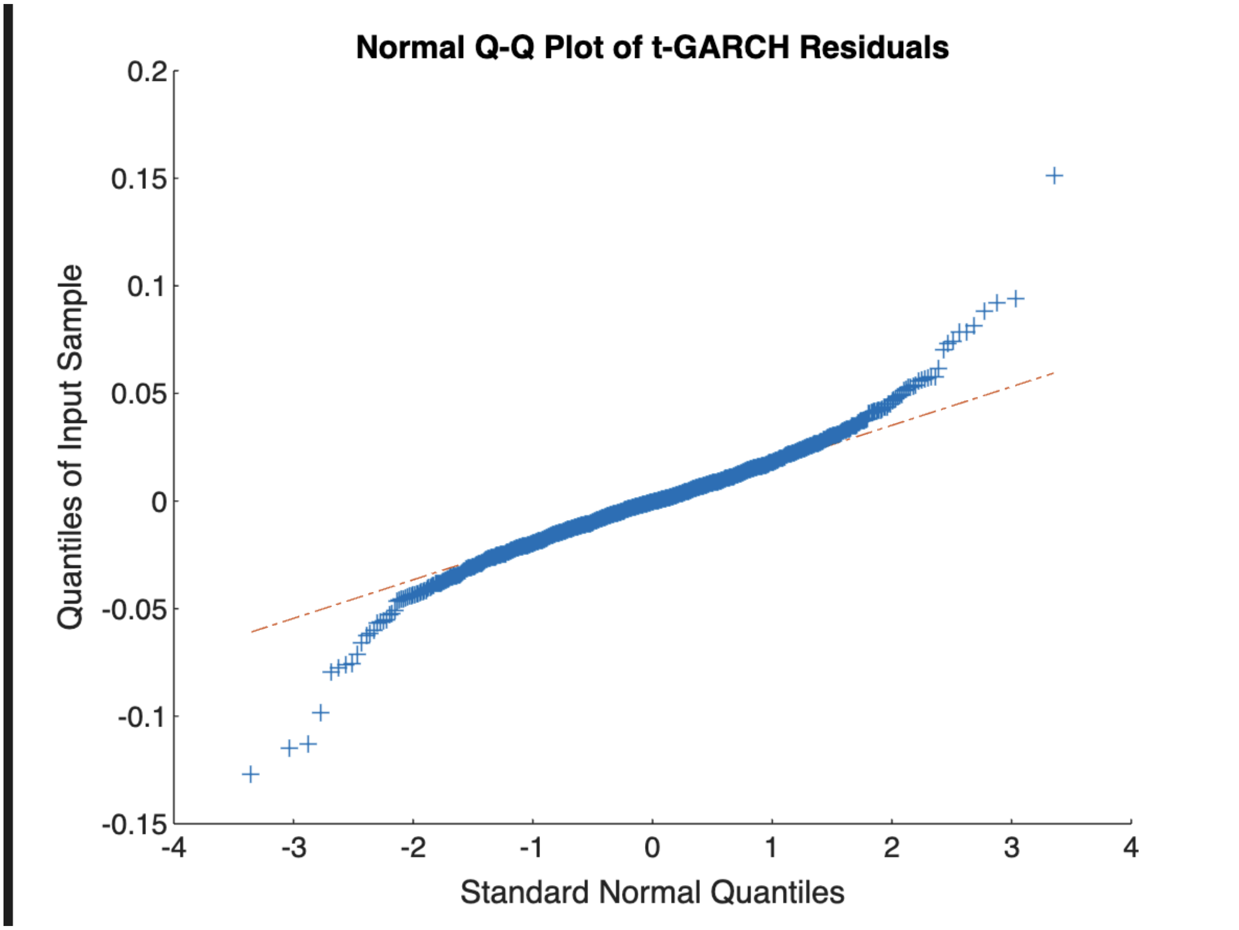
```
% Normal Q-Q Plot
qqplot(timeSeriesData);
title('Normal Q-Q Plot of t-GARCH Residuals');
```

ACF of t-GARCH Residuals:



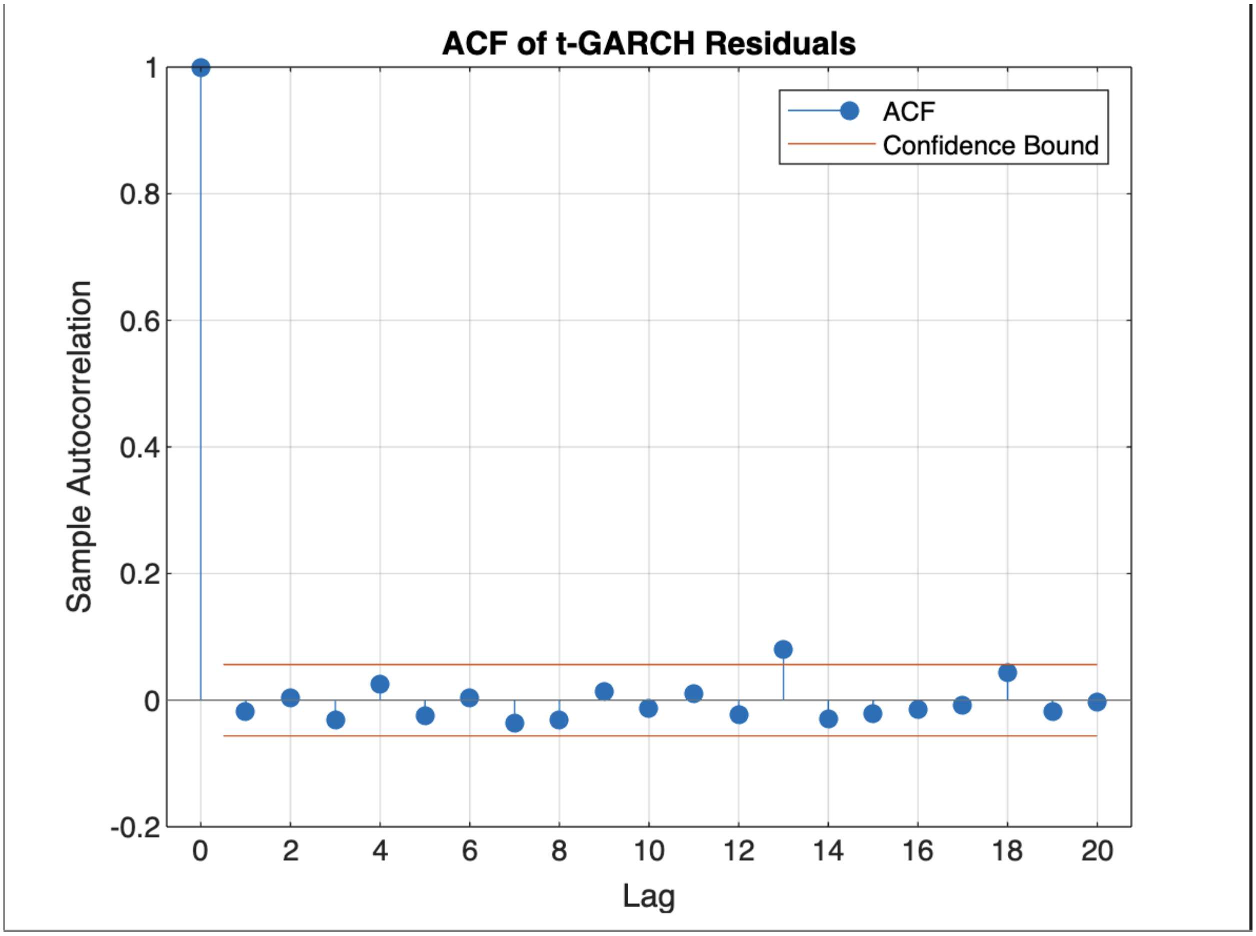
Code:
```
% Autocorrelation of residuals
figure;
autocorr(timeSeriesData);
title('ACF of t-GARCH Residuals');
```

Normal Q-Q Plot of T-GARCH Residual (based of logReturns):



Code:
% Normal Q-Q Plot
qqplot(logReturns);
title('Normal Q-Q Plot of t-GARCH Residuals');

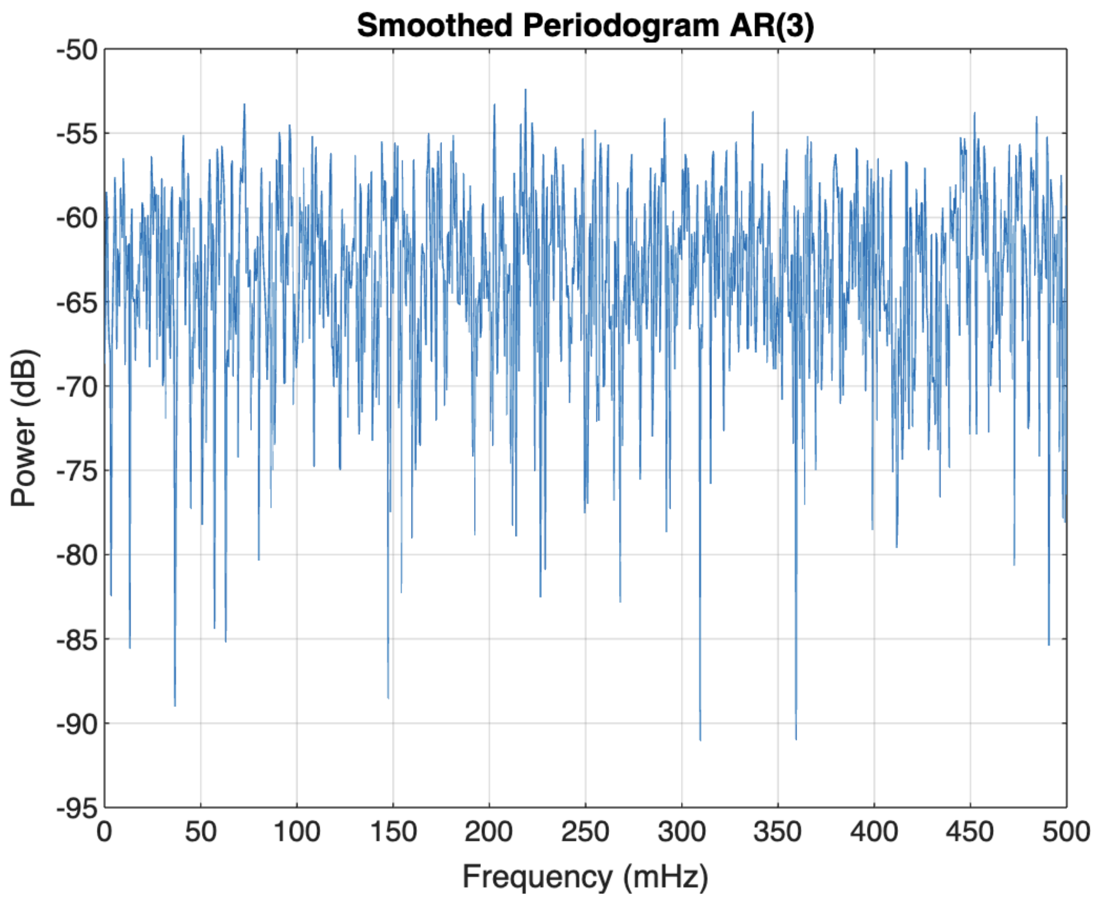
ACF of t-GARCH Residuals (based of logReturns):



Code:

```
figure;
autocorr(logReturns);
title('ACF of t-GARCH Residuals');
```

Smoothed Periodogram:



Code:
```
figure;
periodogram(logReturns,[],[],[], 'power');
title('Smoothed Periodogram AR(3)');
```

AIC Table for Model Comparison:

Model	AIC
{'ARMA(1,1)'}	-5957.8
{'ARMA(2,1)'}	-5956
{'ARMA(2,2)'}	-5954.6

Code:
```
models = {arima(1,0,1), arima(2,0,1), arima(2,0,2)};
AIC_values = zeros(length(models),1);
```

```
for i = 1:length(models)
    fit = estimate(models{i}, logReturns, 'Display', 'off');
    [~,~,logL] = infer(fit, logReturns);
    numParams = length(fit.Constant) + length(fit.AR) + length(fit.MA) + 1; % +1 for
Variance
    AIC_values(i) = aicbic(logL, numParams);
end
T = table({'ARMA(1,1)';'ARMA(2,1)';'ARMA(2,2)'}, AIC_values, 'VariableNames',
{'Model', 'AIC'});
disp(T);
```

Model Fitting (ARMA(2,1)):

	Value	StandardError	TStatistic	PValue
Constant	-0.00060163	0.0010421	-0.57734	0.56371
AR{1}	-0.65089	0.46805	-1.3906	0.16434
AR{2}	0.012011	0.031444	0.38198	0.70247
MA{1}	0.63415	0.46969	1.3502	0.17697
Variance	0.00050464	1.1274e-05	44.761	0

Code:

```
model_21 = arima(2,0,1);
fit_21 = estimate(model_21, logReturns);
```

Inverse Roots ARMA(2,1) - Didn’t work

Inverse Roots ARMA(2,2):

ARIMA(2,0,2) Model (Gaussian Distribution):

	Value	StandardError	TStatistic	PValue
Constant	-0.00012372	0.00030499	-0.40565	0.685
AR{1}	0.063939	0.60663	0.1054	0.91606
AR{2}	0.61128	0.36378	1.6804	0.092883
MA{1}	-0.096057	0.61115	-0.15717	0.87511
MA{2}	-0.60483	0.37159	-1.6277	0.10359
Variance	0.00050441	1.123e-05	44.915	0

Code:

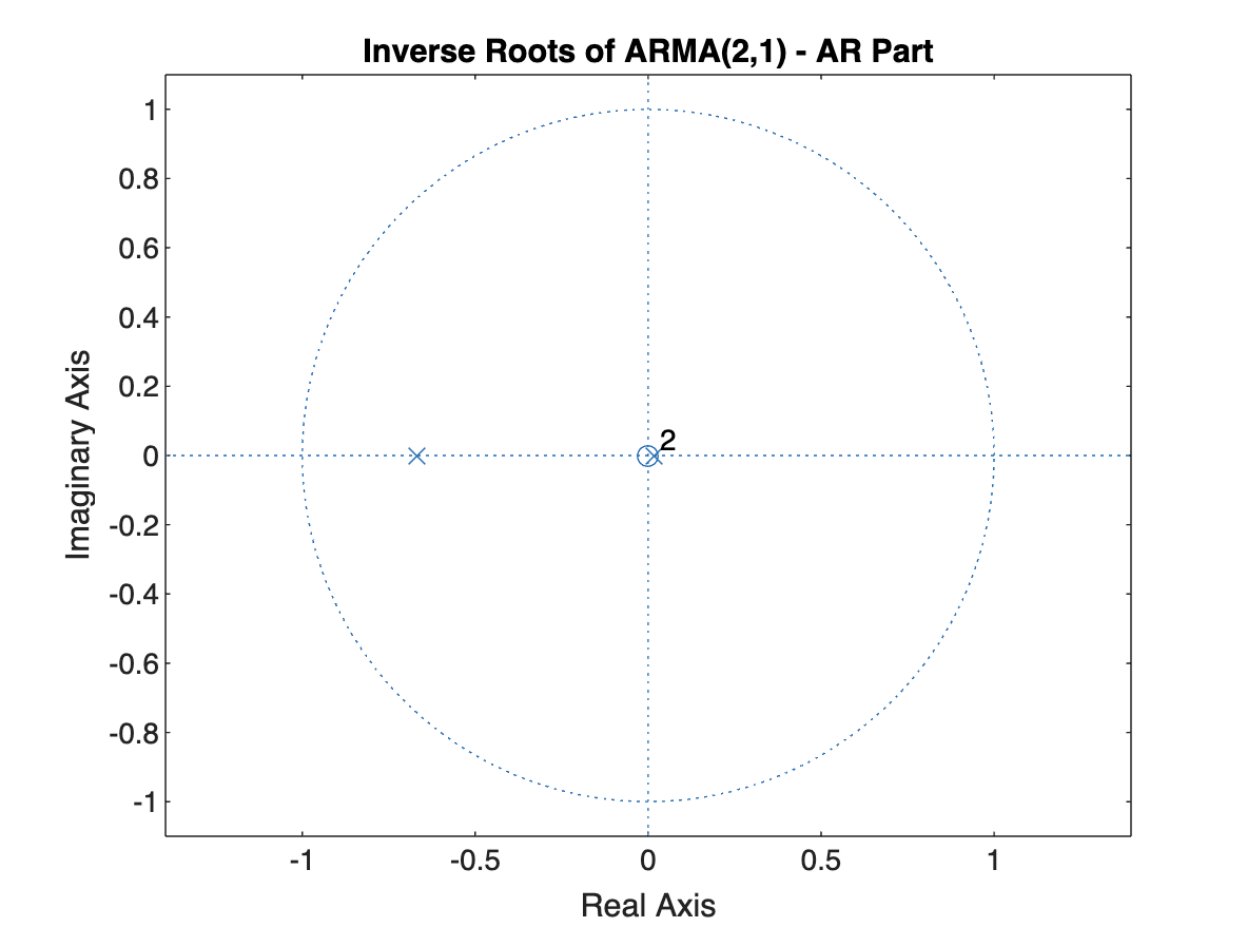
```
model_22 = arima(2,0,2);
fit_22 = estimate(model_22, logReturns);
figure;
zplane(1, [1 -fit_22.AR{:}]);
title('Inverse Roots of ARMA(2,2)');
```

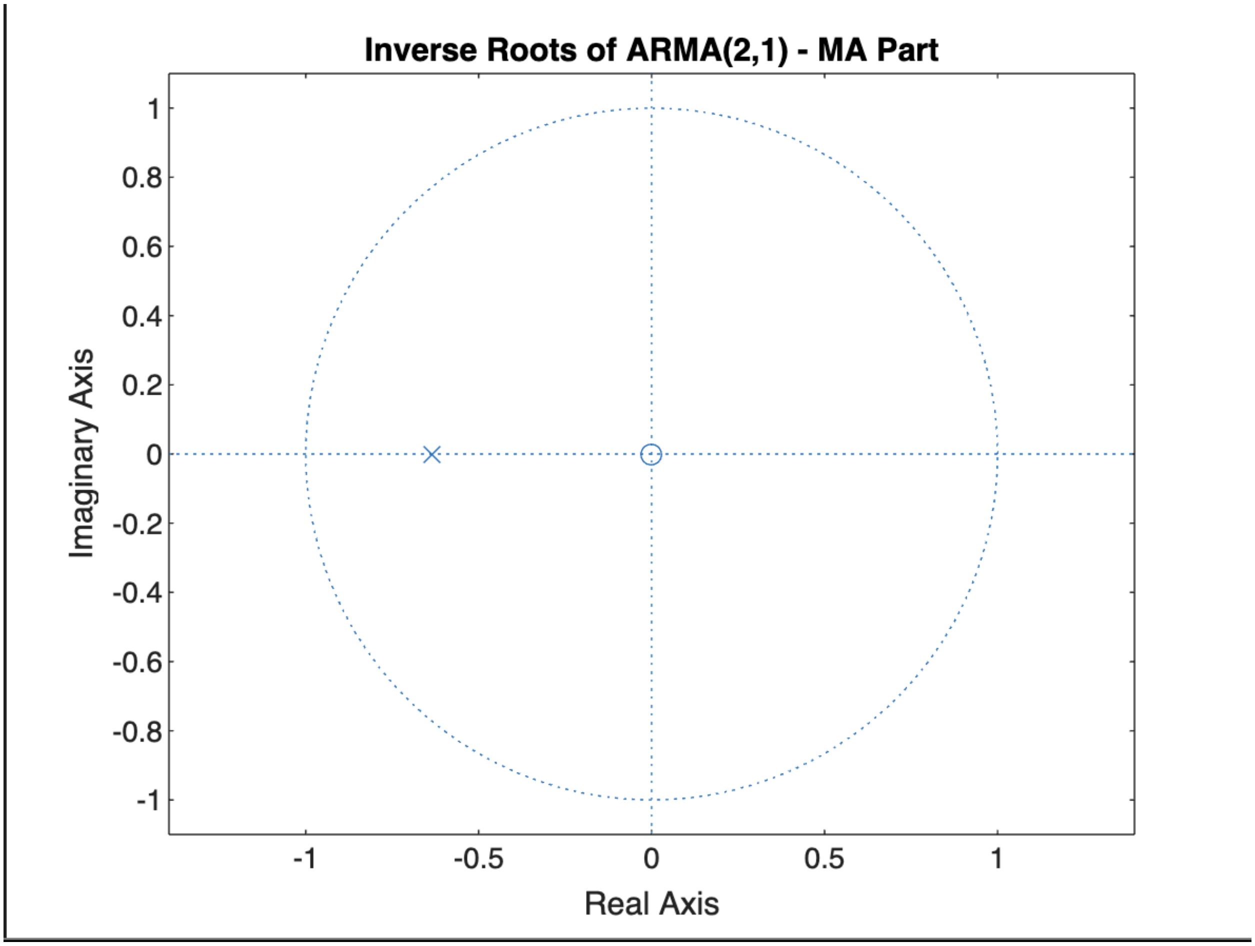
Likelihood Ratio Test between ARMA(2,2) and ARMA(2,1)L
Likelihood Ratio Test p-value: 0.4492

Code:

```
~,~,logL_22] = infer(fit_22, logReturns);
[~,~,logL_21] = infer(fit_21, logReturns);
LR_stat = -2*(logL_21 - logL_22);
p_val = 1 - chi2cdf(LR_stat, 1);  % 1 additional MA parameter
fprintf('Likelihood Ratio Test p-value: %.4f\n', p_val);
```

Inverse Roots of ARMA(2,1) - AR Part and MA Part:

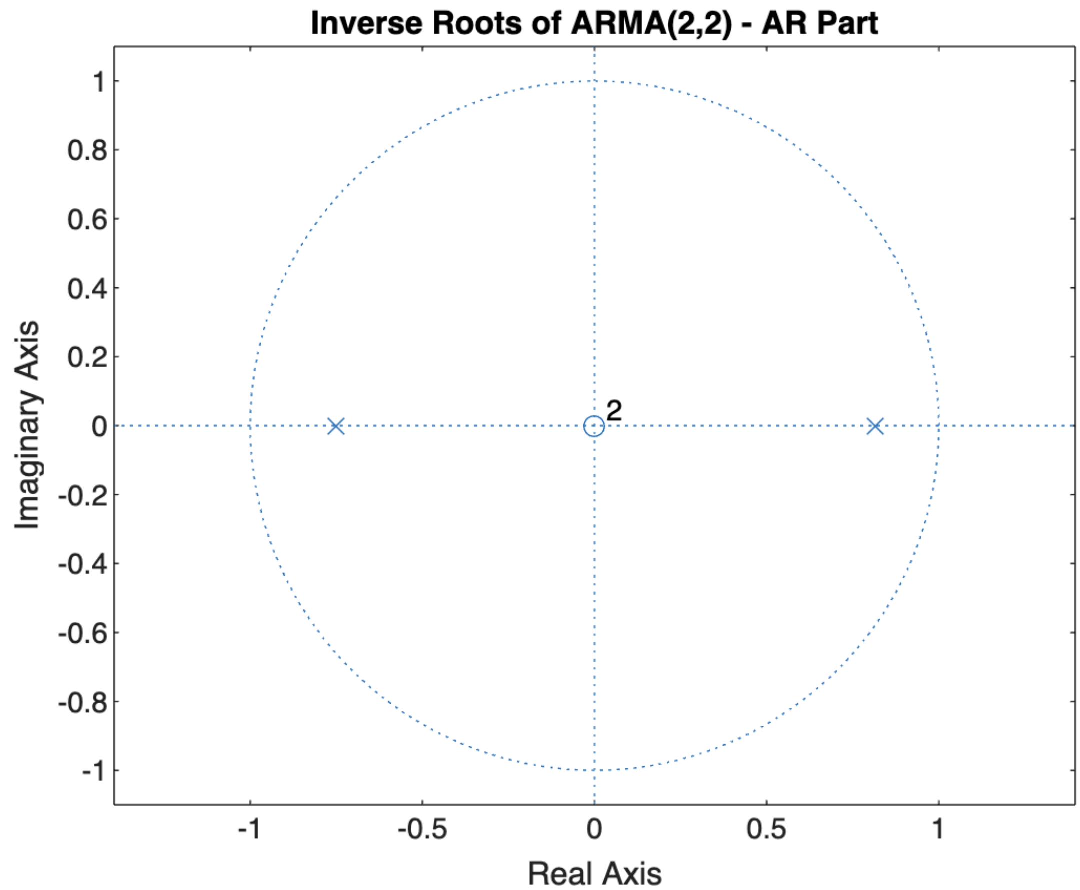


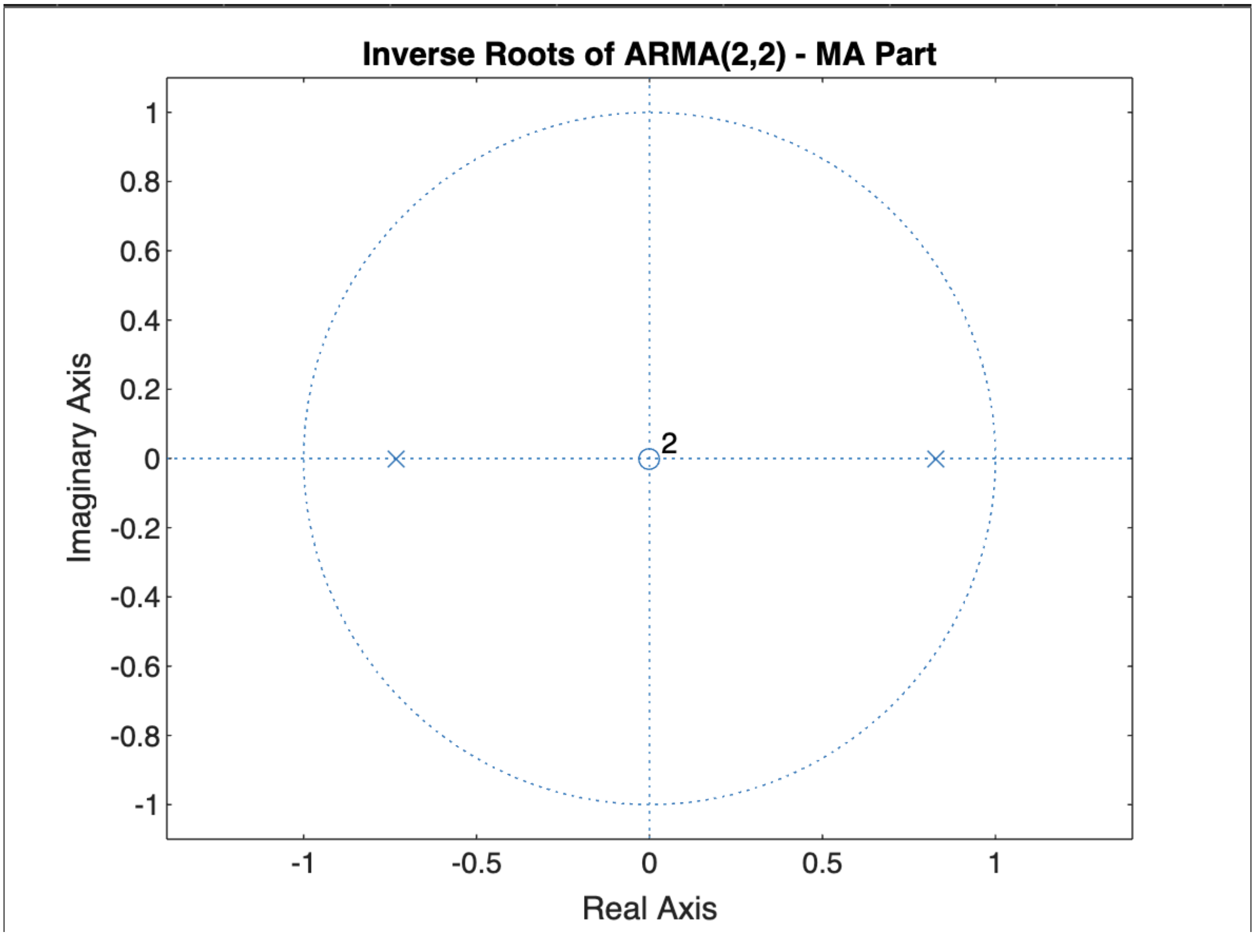


Code:

```
% Assume logReturns is already computed
model_21 = arima(2, 0, 1);
fit_21 = estimate(model_21, logReturns, 'Display', 'off');

% Plot AR roots (for stationarity)
figure;
zplane(1, [1, -cell2mat(fit_21.AR)]);  % AR polynomial
title('Inverse Roots of ARMA(2,1) - AR Part');
xlabel('Real Axis'); ylabel('Imaginary Axis');

% Plot MA roots (for invertibility)
figure;
zplane(1, [1, cell2mat(fit_21.MA)]);   % MA polynomial
title('Inverse Roots of ARMA(2,1) - MA Part');
xlabel('Real Axis'); ylabel('Imaginary Axis');
```

Inverse Roots of ARMA(2,2) - AR and MA Part:





Code:

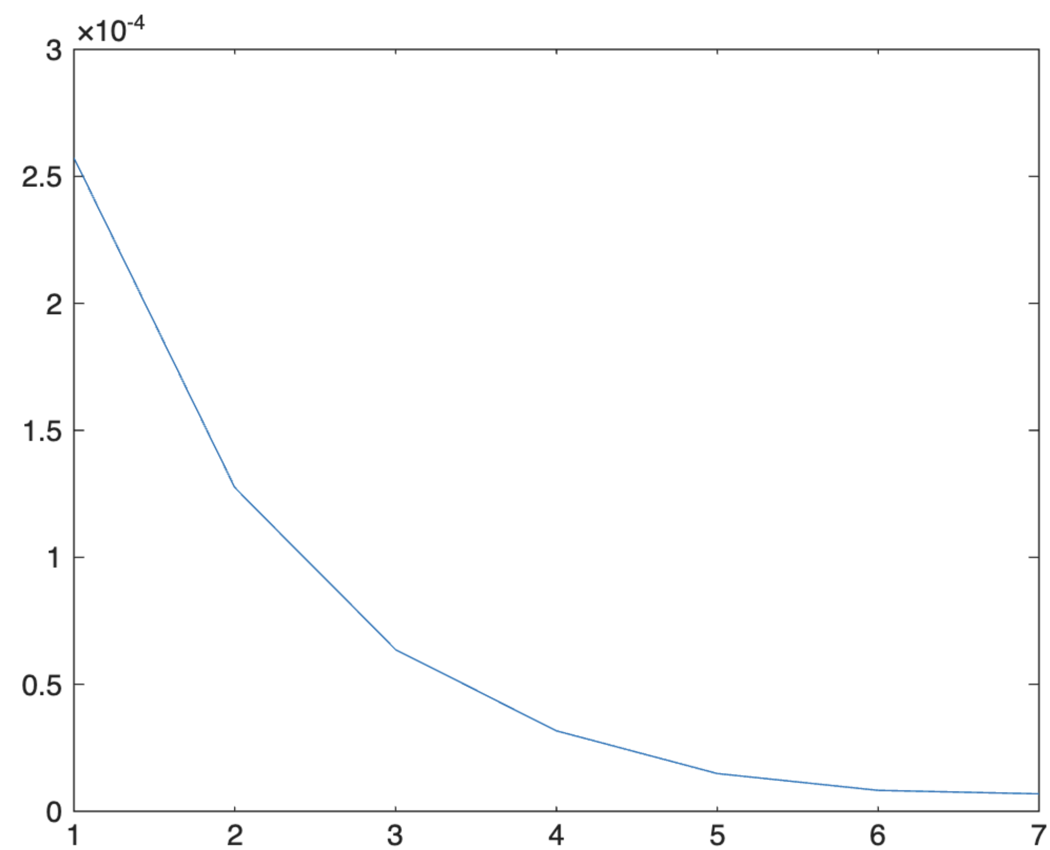
```
model_22 = arima(2, 0, 2);
fit_22 = estimate(model_22, logReturns, 'Display', 'off');

% Plot AR roots
figure;
zplane(1, [1, -cell2mat(fit_22.AR)]);
title('Inverse Roots of ARMA(2,2) - AR Part');
xlabel('Real Axis'); ylabel('Imaginary Axis');

% Plot MA roots
figure;
zplane(1, [1, cell2mat(fit_22.MA)]);
title('Inverse Roots of ARMA(2,2) - MA Part');
xlabel('Real Axis'); ylabel('Imaginary Axis');
```

EGARCH(1,1) Conditional Variance Model (Gaussian Distribution):

	Value	StandardError	TStatistic	PValue
Constant	-0.032444	0.017832	-1.8194	0.068844
GARCH{1}	0.99538	0.0023858	417.21	0
ARCH{1}	0.061922	0.0075463	8.2056	2.2941e-16
Leverage{1}	-0.027469	0.0095519	-2.8758	0.0040305

Code:
model = egarch(1,1);
fit = estimate(model, logReturns);

Wavelet-Based Volatility Analysis:



Code:

```
wt = modwt(logReturns);
plot(modwtvar(wt));  % visualize scale-dependent variance
```